White Paper

Bridging Public Dissemination and Protected Data: A Least-Privilege SDMx Architecture on Azure Databricks

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Classification: Public Reference Architecture
Standards: SDMx 3.0 | BIS Locational Banking Statistics (LBS) | Azure Databricks Unity Catalog
Implementation status: Synthetic provisioning, live control tests and workload teardown completed on Azure. The 6 October 2026 reference evaluation measured approximately 42 minutes to provision with prerequisites already in place and US$10 or less in Azure charges for the deploy/test/teardown cycle; the latest teardown (12 October 2026) took approximately 32 minutes. These are evaluation results, not production service levels; see measurement scope.

> **Independent Reference Architecture Notice:**
> This publication and associated reference implementations were developed in a personal capacity using synthetic data fixtures and publicly available international statistical standards (SDMx 3.0, BIS Locational Banking Statistics). They are shared for educational purposes and reflect only the author's views. This work is not affiliated with, sponsored by, or representative of the Bank of Canada, the Federal Reserve System, the Bank for International Settlements, or any official statistical institution.

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1. Executive Summary & The Contractor Dilemma

International financial institutions, central banks and national statistical offices face a structural conflict between two fundamental institutional mandates:

1. **Public Statistical Transparency:** The imperative to disseminate macroeconomic and financial indicators openly to researchers, market participants, and member states.
2. **Confidential Statistical Protection:** The obligation to safeguard restricted observations, jurisdictional entitlements and the confidentiality of contributing institutions.

A synthetic-first delivery model enables external specialists to implement and test a statistical platform against approved structures and security boundaries without access to confidential production records. That matters because the engineering is increasingly delivered by systems integrators, contractors and AI coding agents, while statutory confidentiality forbids showing them the data the controls protect. SovereignShield provides an Azure and Databricks reference implementation alongside established SDMx infrastructure and institutional controls.

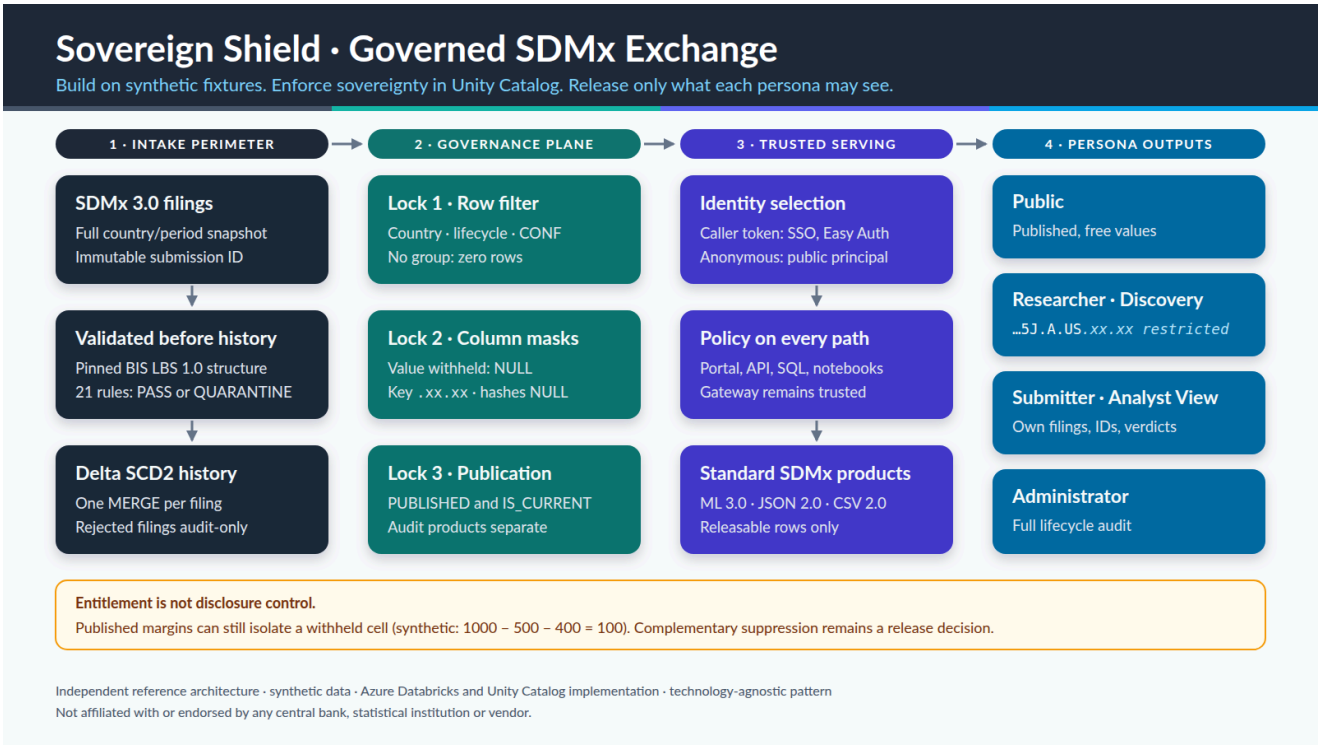


Figure 1 — Executive Architecture: intake perimeter, Unity Catalog governance plane, trusted serving plane and persona outputs, including the researcher Discovery Gateway.

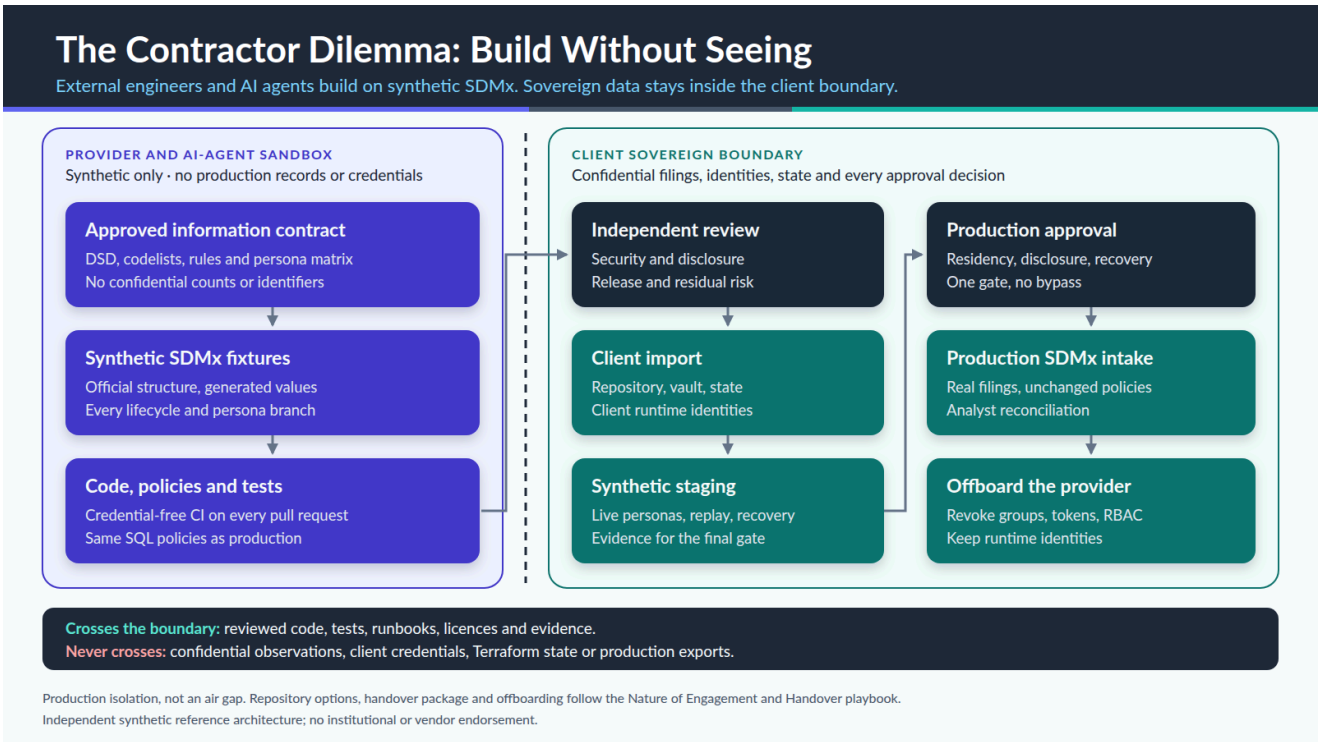


Figure 2 — The Contractor Dilemma: providers and AI agents build in a synthetic-only sandbox; confidential filings, identities and every approval stay inside the client boundary. The boundary is production isolation, not an air gap.

SovereignShield combines SDMx statistical contracts, submission-aware history and query-time entitlement enforcement. Azure Databricks Unity Catalog implements the policy controls in the demonstrated deployment.

Specialists implement against an approved synthetic contract, and client-controlled deployment identities remain independent of personnel changes. PR verification has no cloud credentials. Privileged planning and deployment require an explicit manual run on reviewed `main` and a protected GitHub environment. Runtime

secrets, state and plans remain protected operational assets.

The companion Executive Brief carries the same title, **Bridging Public Dissemination and Protected Data: A Least-Privilege SDMx Architecture on Azure Databricks**, and summarizes the pattern for decision-makers. The Nature of Engagement and Handover defines delivery responsibilities.

Submission boundary: the modeled international exchange accepts SDMx files, not bank micro-transactions. Synthetic bank micro-transactions exist solely as educational fixtures illustrating how realistic statistical observations and confidentiality flags are calculated. The demo ledger is not an institutional intake requirement or system deliverable. Domestic granular-data collections and other reporting regimes are outside this exchange contract.

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2. The Dual Consumption Model

Statistical platforms support multiple cadences. This reference submission profile demonstrates quarterly (`Q`) LBS data. The pinned BIS frequency list is authoritative: generic stress-fixture cadence labels must not be mistaken for validated LBS codes.

A **Unified Storage, Dual-Tier Consumption Model** provides public and entitled access through Unity Catalog and a decoupled gateway. Separate deployments or storage boundaries may be required where an institution mandates physical residency or stronger administrative isolation.

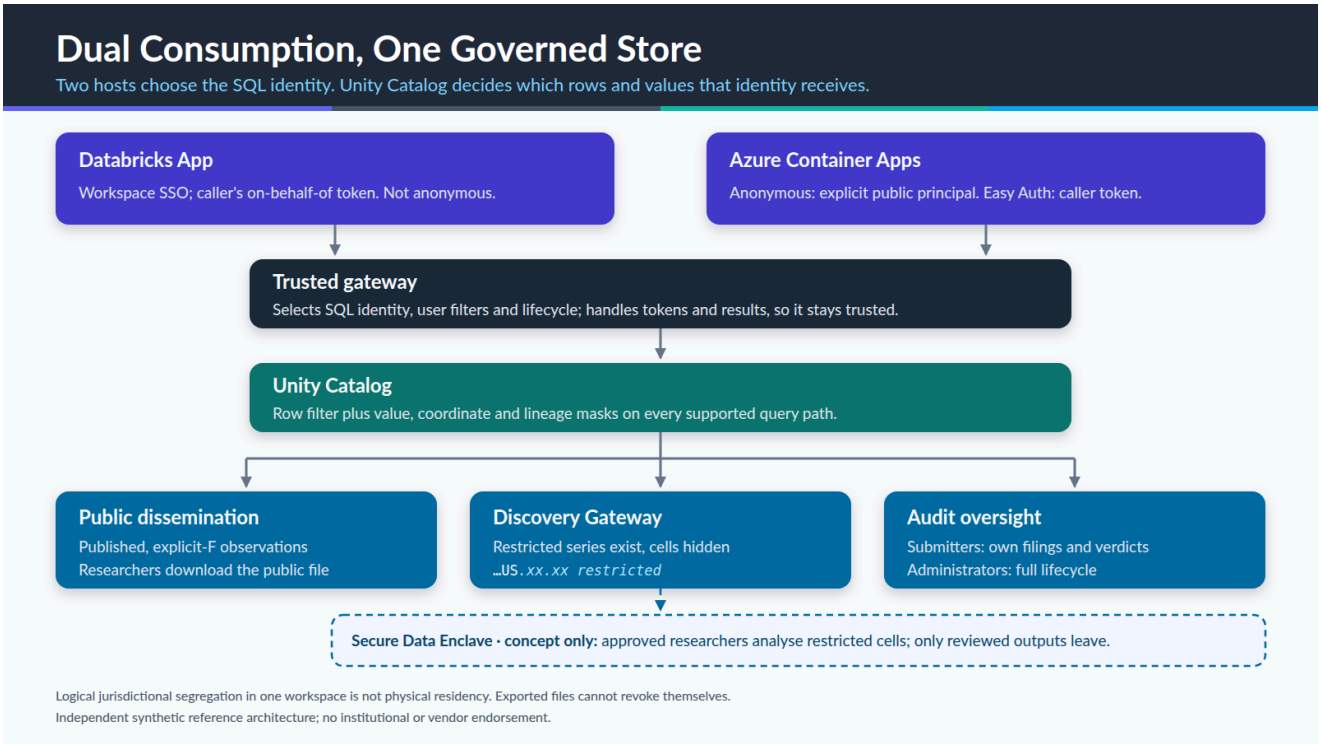


Figure 3 — The Dual Consumption Model: public dissemination, the researcher Discovery Gateway and audit oversight read one governed store. The Secure Data Enclave is a conceptual extension.

The Perimeter Identity Problem

Databricks Apps enforce workspace sign-in, so they cannot by themselves provide a genuinely anonymous public endpoint. Sovereign Shield therefore deploys the same FastAPI portal through two hosts:

- **Databricks App:** signed-in workspace users query with an on-behalf-of SQL token. The caller's Databricks account-group memberships reach Unity Catalog unchanged.
- **Azure Container Apps gateway:** anonymous requests query as `spn-sovereignshield-public`, an explicit service principal that belongs only to `sg-sovereignshield-public`. Optional Entra Easy Auth uses `AllowAnonymous`; after sign-in, the browser forwards the caller's Azure Databricks access token to the same API instead of using the public proxy identity.

The gateway selects the SQL identity and lifecycle query. Unity Catalog applies caller-aware table policies. Standard SDMx exports contain only the current published snapshot; a separate audit CSV preserves rejected filings, submission identity, timestamps, and failure feedback. The gateway remains trusted because it handles bearer tokens and entitled results. A compromised gateway is not harmless.

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3. The Triple-Lock Security Blueprint

At the core of the data plane sits the **Triple-Lock Governance Architecture**, implemented natively within Unity Catalog SQL functions and Entra ID security claims.

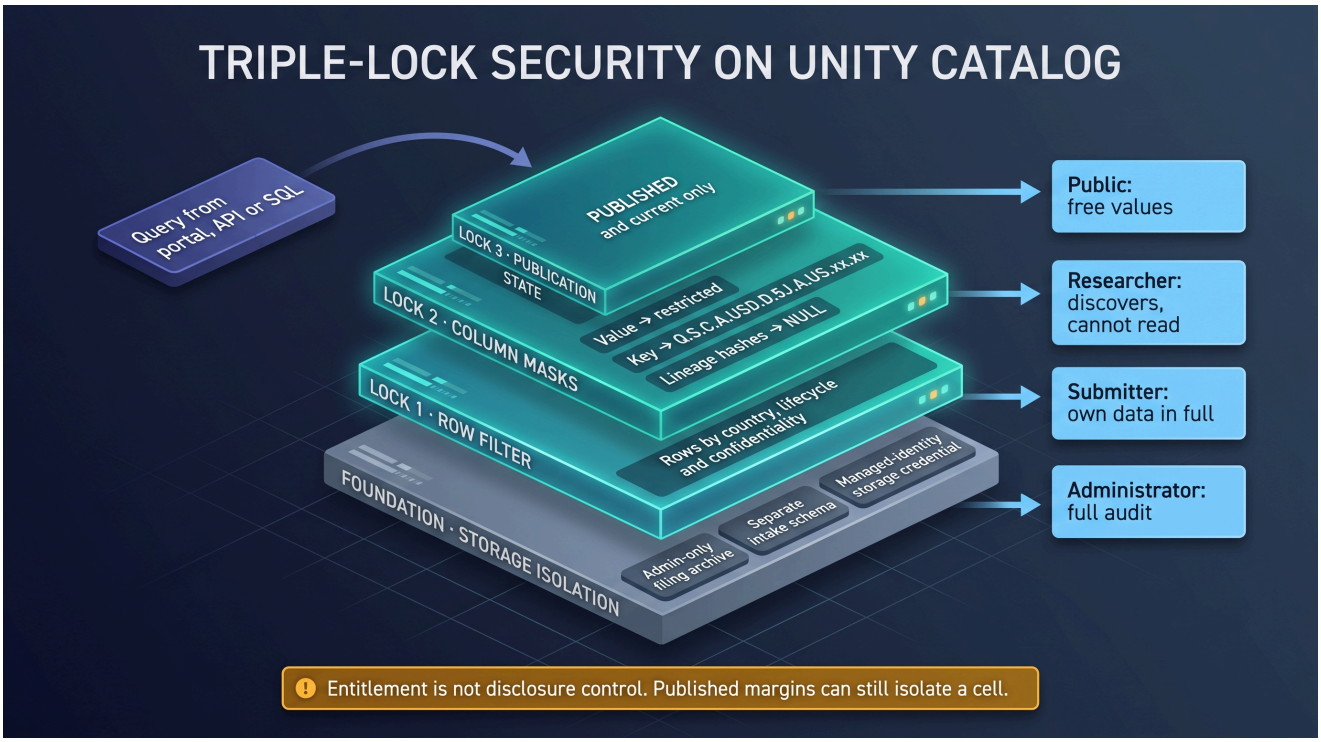


Figure 4 — Triple-Lock Policy Enforcement Point: jurisdictional row filtering; value, coordinate and lineage masking; and publication-state controls. AI-generated illustration; the hand-authored control contract is the reference.

The Persona Matrix

Four entitled roles are evaluated through Databricks account-group membership, with a deny-by-default case. Setup reconciles selected Entra identities; it is not continuous directory synchronization:

1. **Public Consumer (`sg-sovereignshield-public`):** The anonymous browser is not itself a Databricks identity; the Container Apps gateway maps it to the dedicated public service principal. Access is restricted strictly to published observations explicitly marked free for publication (`OBS_CONF = 'F'`). Confidential rows are excluded from the query result by Unity Catalog.
2. **Discovery Gateway Researcher (`sg-sovereignshield-researchers`):** The researcher is not a reader of partially suppressed tables but the user of a *Sovereign Discovery Gateway*. Every published series is listed, so a researcher learns that restricted observations exist for a reporting jurisdiction, period and series family, and can open a bilateral conversation with the submitting central bank or national statistical office. For any observation not explicitly free (F), including unknown or missing classifications, Unity Catalog withholds the value, replaces the counterparty coordinates with `xx.xx` and nulls the lineage columns. Discovery confers no access to the value; restricted cells require the Secure Data Enclave.
3. **Regional Reporting Submitter (`sg-sovereignshield-submitter-{cty}`):** Row-Level Security grants full, unmasked access strictly to observations originating from the submitter's designated jurisdiction, read from segment 9 of the SDMx key. For foreign jurisdictions the submitter inherits the public tier: published, free-to-publish observations only. The **Analyst View** is this submitter workflow, not a separate entitlement. It allows analysts to reconcile the latest filing they expect the

international organization to hold with the submission IDs, timestamps, values and validation outcomes actually held in the system. The latest submitted filing may be rejected; it must not be confused with the current accepted publication.

4. **Central Auditor / Administrator (`sg-sovereignshield-admin`):**
   Full, unmasked access across all jurisdictions and all lifecycle states, including quarantined batches, for regulatory oversight.
5. **No recognised membership — zero rows.**
   Each entitlement requires positive membership. Removing all relevant account-group memberships removes this data entitlement, but complete offboarding also requires Entra and Databricks membership reconciliation, session/token revocation, Azure and GitHub access removal, and ownership review. A person with control-plane or ownership privileges remains a separate trust boundary.

The SQL examples are in Appendix A. The deployment executor creates content-addressed functions, checks their definitions, changes bindings without a detach operation, and verifies binding metadata. It aborts on any error and does not automatically migrate legacy table types. A failed deployment can leave old and new protected bindings on different objects; this is not a cross-object atomic policy migration.

Every policy reads the reporting country from `L_REP_CTY`, an unmasked column the writer copies from segment 9 of the key and checks against it, because Unity Catalog rejects a masked column, such as the coordinate-masked key, as the input of another policy. A missing anchor fails closed. Strict input checks verify canonical keys and pinned codelists before persistence. The mask independently checks the reporting country for own-country access.

Memberships compose additively. A principal who is both a submitter and a researcher receives the union of the matching row entitlements, while the mask still reveals restricted values only for the principal's own jurisdiction. This is why the functions use independent `OR` branches rather than a first-match `CASE` expression.

The Sovereign Discovery Gateway

A withheld value must not carry its exact coordinates. `fn_ddm_series_key_mask` applies the same reveal rule as the value mask: administrators, the owning submitter and explicitly free observations keep the stored key; for everyone else segments 10 and 11, `L_CP_SECTOR` and `L_CP_COUNTRY`, become `xx`. `Q.S.C.A.USD.D.5J.A.US.A.5J` is shown as `Q.S.C.A.USD.D.5J.A.US.xx.xx`, and the portal renders the suffix in the same muted italic style as the word *restricted*.

Four properties make this more than presentation. Segments 1-9 are preserved, so a masked key still names the reporting country held in `L_REP_CTY`, the unmasked sovereignty anchor every policy reads. Masks resolve before query predicates, so filtering on a counterparty cannot probe which restricted row it belongs to, and filter lists never offer codes held only by restricted rows. `RECORD_ID` hashes the full key with a visible submission ID, `version_hash` hashes the value with visible attributes and `VALIDATION_NOTES` names the rules a row takes part in; `fn_ddm_lineage_mask` nulls all three, closing a key-enumeration and value-confirmation side channel. Finally, coordinate-masked rows are discovery metadata, never data: every download excludes them before its row limit applies, so the researcher's product equals the public one and the two cannot be differenced.

Coordinate masking hides *which* counterparty a restricted cell describes. It does not hide the residual of a published margin across dimensions that remain visible. In the fixture, the CA claims total (1000), domestic currency (400) and foreign currency (500) are public, so the unallocated-currency cell is still $1000-400-500=100$ even though its counterparty coordinates are masked. The fixture has a single restricted cell; when several cells in the same table are masked, as is typical for most statistics, working back to any one value is much harder, though not necessarily impossible. What is needed beyond the implemented controls, such as complementary suppression of a margin or withholding the role, is therefore the disclosure authority's case-by-case decision, based on an understanding of the statistic the architecture is applied to.

The coordinate and lineage masks are verified offline, against the policy SQL contract and the local mirror. Their live Unity Catalog acceptance is a pending release gate.

Demonstrated Persona Outcomes

Figures 5, 7 and 8 are synthetic portal captures supplied on 18 September 2026, with their captured labels, values and filters unchanged. The researcher capture is a local synthetic-fixture capture from 1 October 2026 that shows the Discovery Gateway; its persona is a labelled browser fixture over the local policy mirror, not live SSO. The capture inventory records each displayed state; screenshots alone are not independent identity or disclosure-control acceptance evidence.

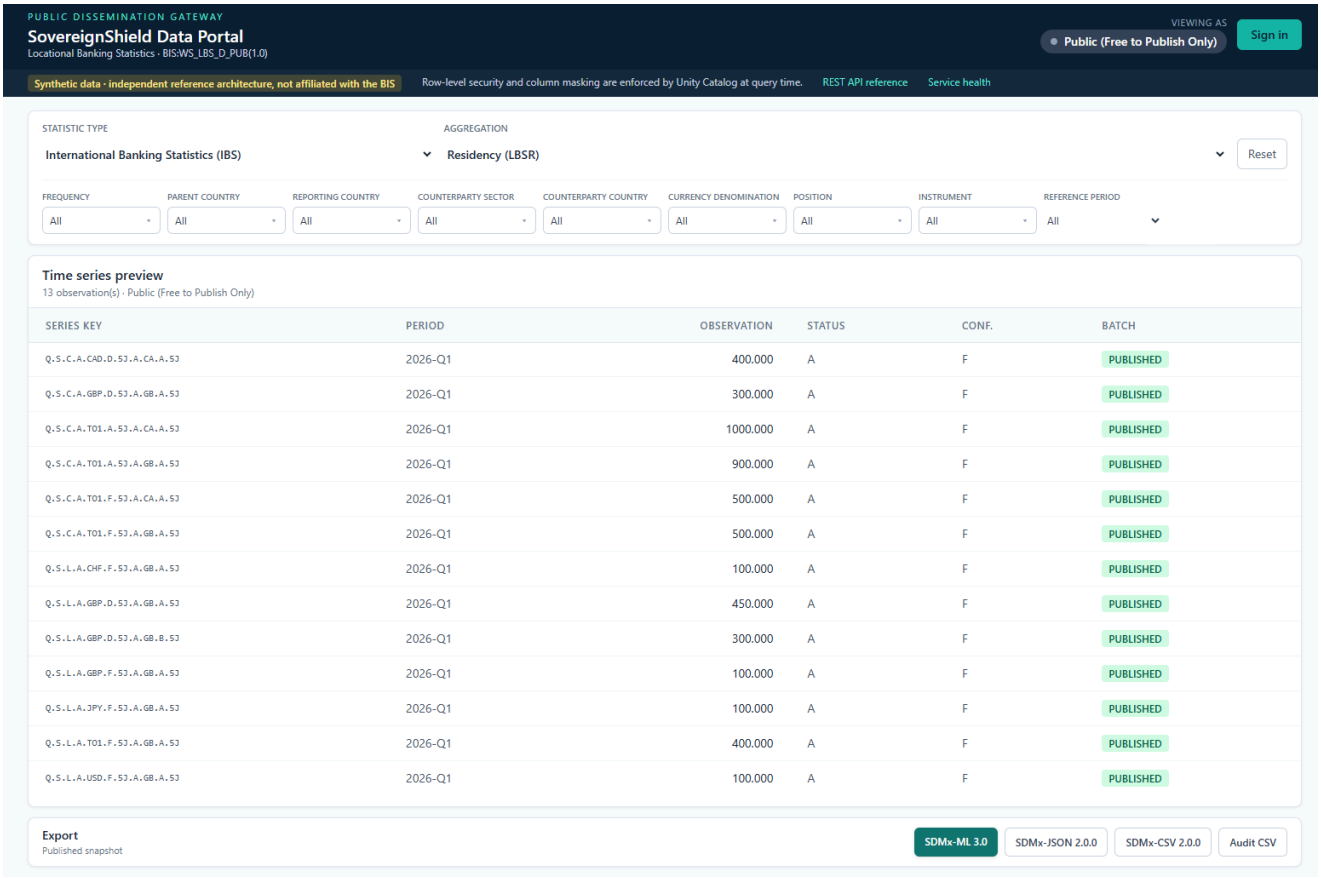


Figure 5 — Anonymous Public Portal View: the explicit public proxy identity receives 13 published, free-to-publish observations.

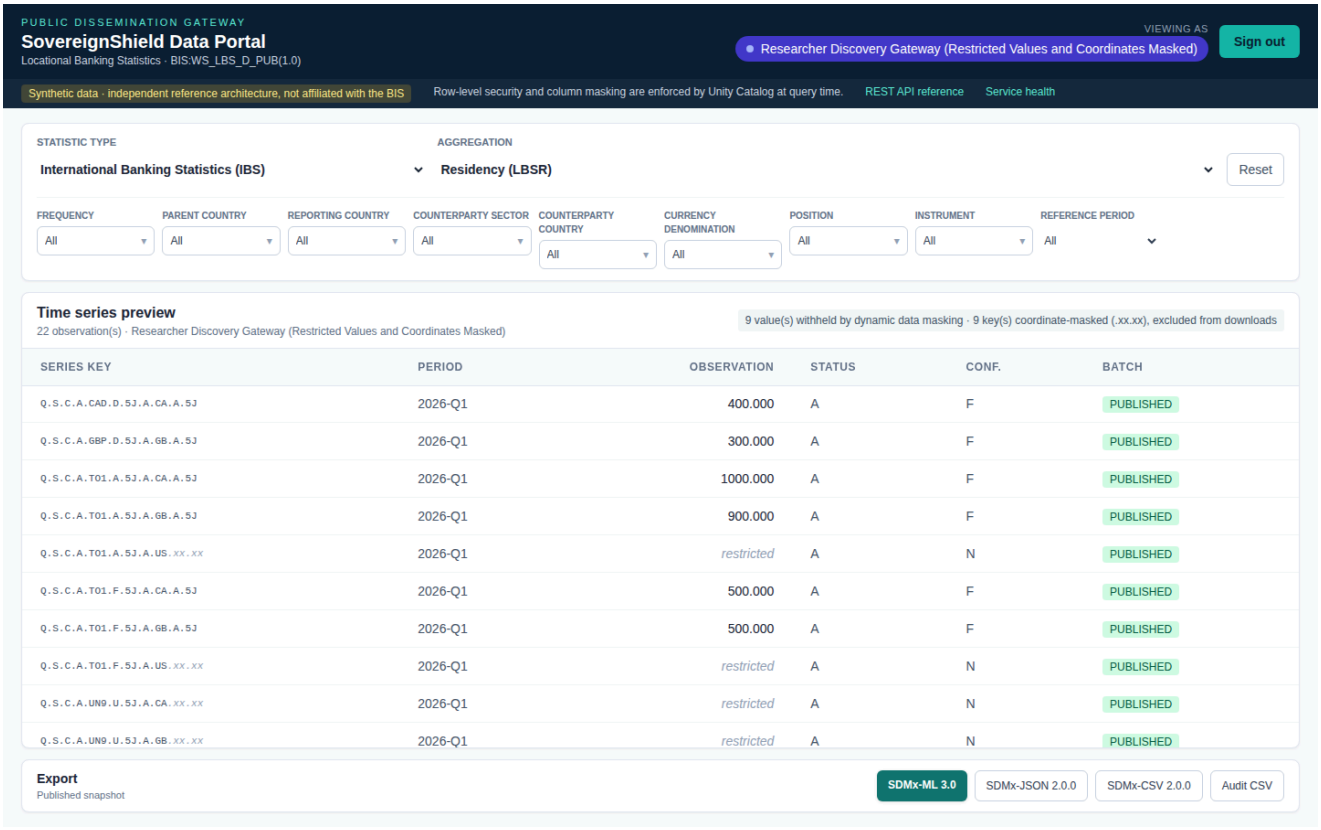


Figure 6 — Researcher Discovery Gateway: 22 published observations; nine restricted values are withheld and their counterparty coordinates shown as `.xx.xx` in the same muted style. These rows are excluded from every download.

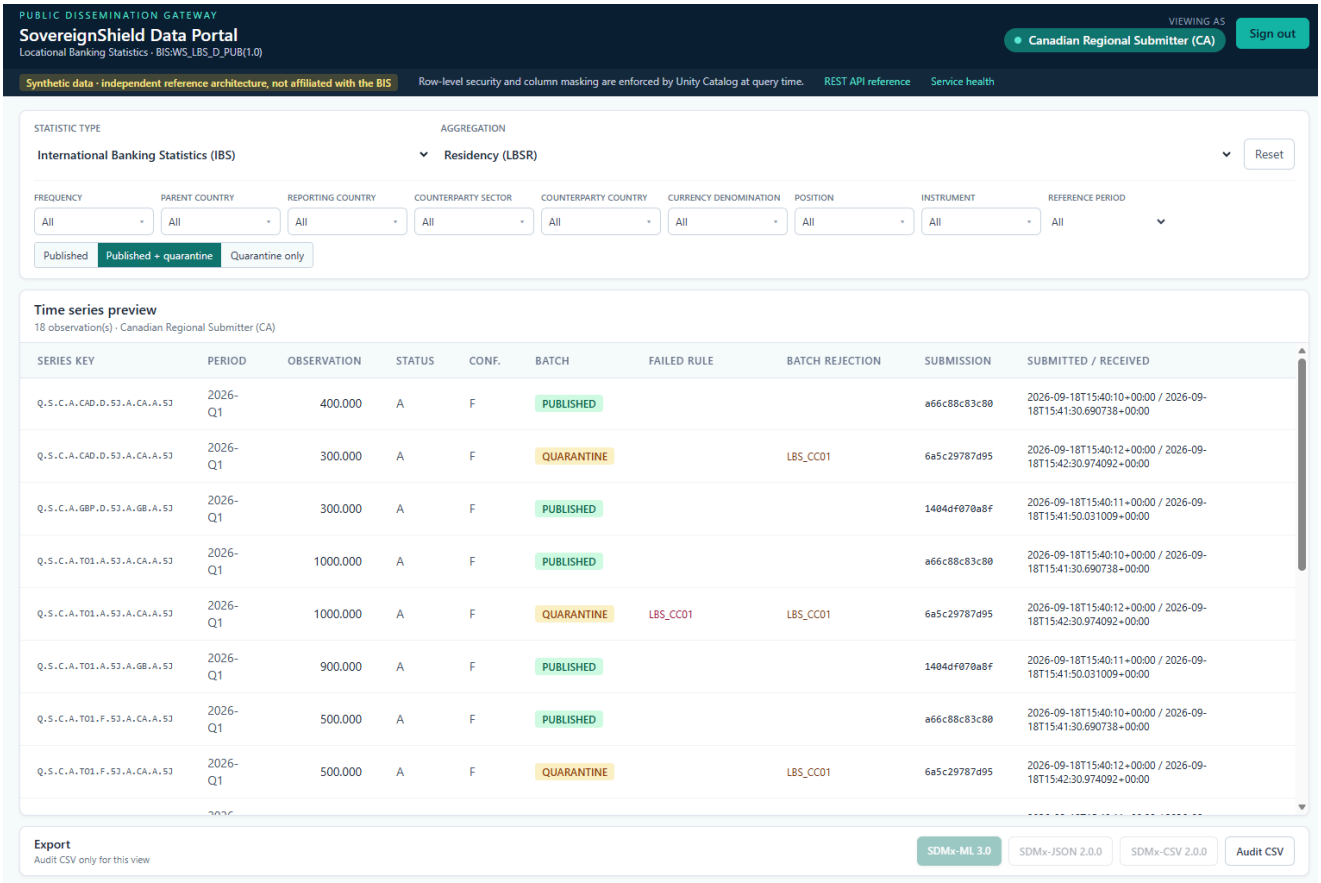


Figure 7 — Analyst Reconciliation View: the Canadian Regional Submitter (CA) sees 18 observations in Published + quarantine mode, with submission identifiers, submitted/received timestamps and rejection feedback. Own filings remain distinguishable from foreign public observations.

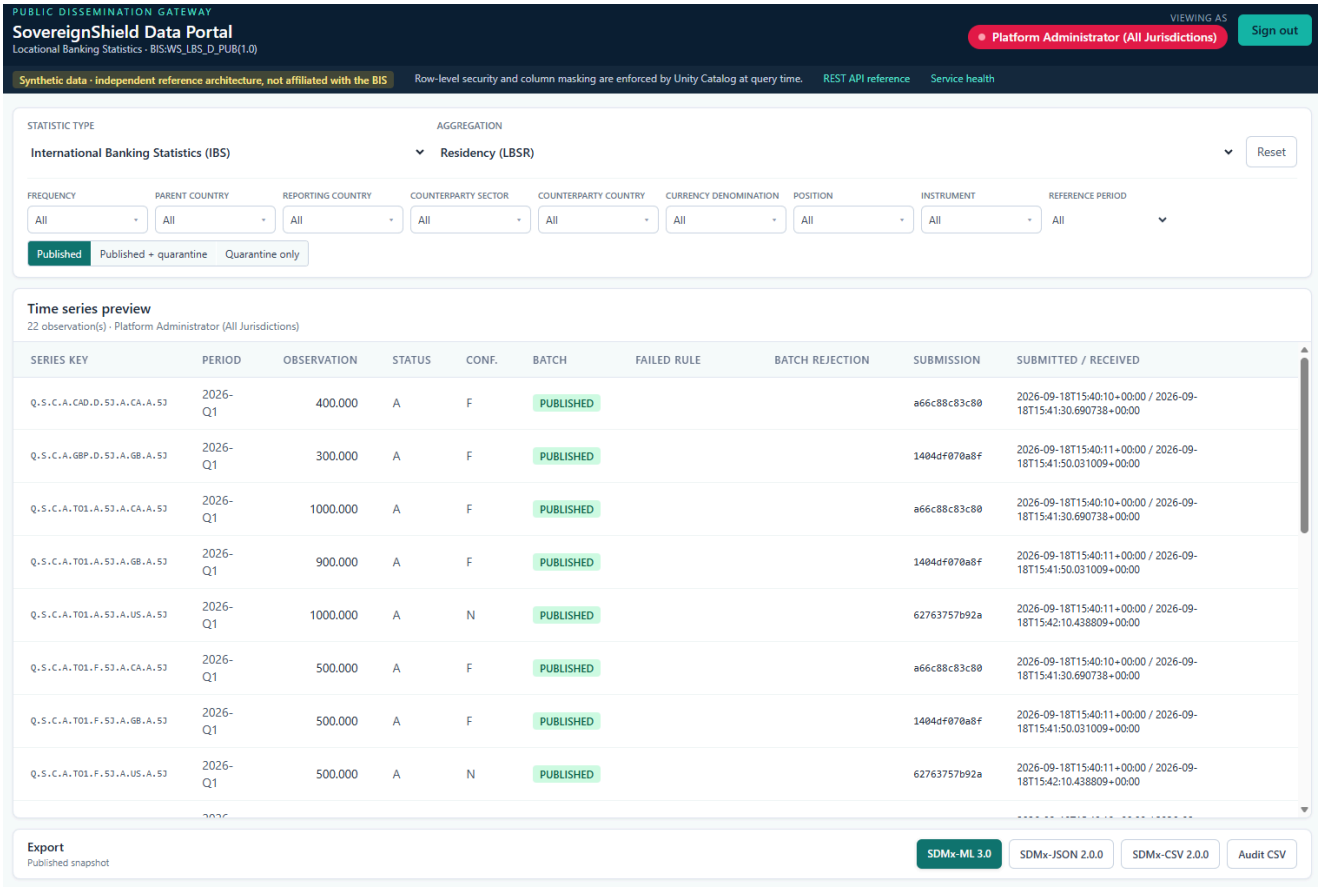


Figure 8 — Administrator Published View: the captured Published mode contains 22 observations, including restricted measures visible to the administrator. This image does not show the separate 44-row Published + quarantine mode.

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4. Declarative Governance: Separation of Concerns

Infrastructure provisioning and data-plane policy have separate ownership to prevent competing writers and inconsistent deployment state.

- **Terraform owns the Infrastructure Control Plane:** Entra groups and deployment identities, Azure resources, the Databricks workspace, Unity Catalog storage credentials and external locations, catalogs, schemas, SQL warehouses, and grants. The workspace resolves its existing regional metastore attachment through the Databricks provider; this configuration does not create or bind an account-level metastore. Terraform never manages table DDL, row-filter bindings, or column-mask bindings.
- **Databricks Asset Bundles (DABs) & SQL own the Data & Policy Plane:** Table DDL, policy UDF logic, row filter attachments (`SET ROW FILTER`), and column mask attachments (`SET MASK`) are version-controlled alongside pipeline logic in `unity_catalog_triple_lock.sql`.
- **Secret Handling:** CI federation avoids a stored deployment client secret. The public SQL proxy still uses a secret resolved from Key Vault; managed identity retrieves that reference, not all downstream SQL credentials. Terraform state and saved plans contain sensitive credential values and require restricted storage, encryption, retention, and auditing. The 90-day rotation resource acts on a subsequent apply; consumer refresh must be verified. There is no independent scheduled rotation service in this release.

Technology-Agnostic Pattern, Platform-Specific Controls

The core framework is technology-agnostic: versioned SDMx contracts, explicit entitlements, synthetic-first delivery, atomic submission transitions, separate publication/audit products and client-owned runtime identities. Azure and Databricks are the demonstrated implementation, not a requirement of those principles.

Terraform's provider and module boundaries support extension to AWS, GCP, Microsoft Fabric or an open-source software combination. Such extensions require platform-specific identity, storage, query-policy, transactional-history and deployment adapters. Unity Catalog functions, Databricks Apps and Azure authentication are not portable unchanged. Each implementation must pass equivalent persona, disclosure, lifecycle and recovery tests; Terraform does not establish control equivalence by itself.

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5. Temporal Integrity & SDMx Compliance

Statistical reporting data is non-destructive; retrospective revisions are common as member institutions re-evaluate balance sheet exposure. A robust platform must maintain a complete historical audit trail without breaking downstream analytics.

Three Data Zones, Three Sensitivity Boundaries

The catalog is divided by what each object represents and who can safely traverse it:

Schema	Contents	Access boundary
`sovereign_submissions`	Governed SDMx-ML filing archive; the demonstration also retains educational synthetic micro files	Administrator only; volumes cannot carry row filters or column masks
`sovereign_intake`	Educational synthetic bank micro-transaction ledger illustrating aggregation and classification; not international intake	Administrator plus reporting submitters in the demo; `fn_rls_micro_country_lock` restricts each submitter to its own country
`sovereign_shield`	Macro SCD2 history, policy functions, and published view	All recognised personas can traverse; multi-column RLS and DDM determine rows and values

This separation prevents a broad schema grant intended for disseminated aggregates from accidentally making raw files or institution-identifying rows reachable. The public portal and researcher persona never receive access to the submission volume or micro ledger.

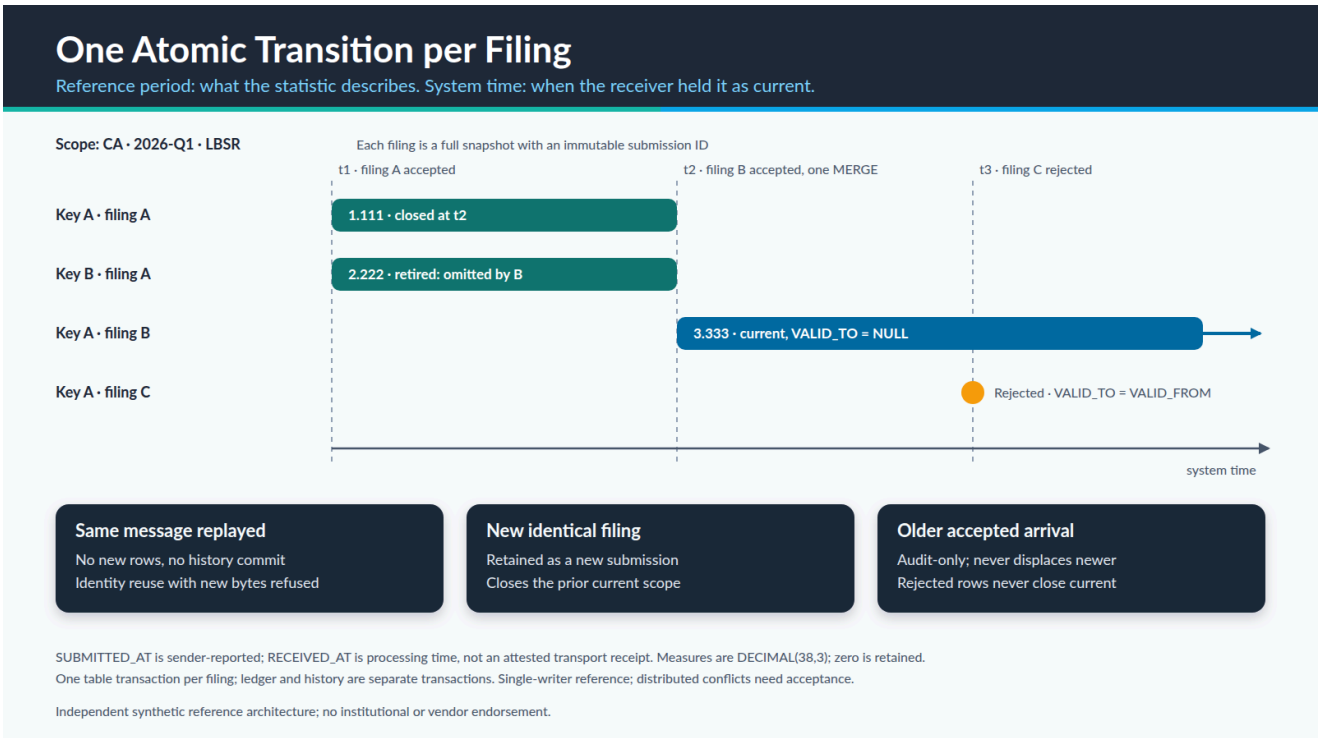


Figure 9 — Temporal SCD2 Delta Merge: accepted revisions update the current state; rejected revisions remain audit-only and preserve the last accepted observation.

- **Atomic Submission Transition:** Accepted expiry and insertion occur in one Delta `MERGE`, using `VALID_FROM`, `VALID_TO`, and `IS_CURRENT`. Local Delta and live Databricks acceptance tests cover replay without row duplication and smaller full replacements. A new identical filing retains a distinct submission identity. The Spark job is single-writer; distributed concurrency and production throughput require separate acceptance.
- **Two Independent Verdicts:** The reporting authority owns confidentiality classification; the receiving organization validates the submitted SDMx file independently. The synthetic generator illustrates classification with a configurable 0.60 dominance threshold. That educational rule is neither a universal reporting requirement nor a complete disclosure-control method. A confidential observation can be arithmetically valid.
- **The Submission Is the Contract:** The reporting task authors an SDMx-ML 3.0 file per jurisdiction per cycle; the receiving task validates its contents. It does not recompute the submission from bank micro-transactions. The separate synthetic ledger only explains fixture construction and demonstrates an additional access boundary.
- **Rules Read, Coverage Stated:** The validator parses 21 within-dataset arithmetic rules from the workbook. Six further cross-collection entries (`LBS_CC:22` through `:27`) are explicitly reported as not implemented. Missing breakdowns are reported as not evaluated. The workbook interpreter, including its `ISO` placeholder handling, is an implementation that needs independent domain review; parsing a rule is not proof of its complete semantics.
- **Atomic Verdict, Precise Attribution:** Acceptance is scoped to a full `(reporting country, reporting period, aggregation)` submission. A failed submission is withheld as a batch. `FAILED_RULE_ID` identifies the checks failed by each observation; `BATCH_FAILED_RULE_ID` records the submission-wide reason. Analysts can distinguish rejected arrivals from the accepted data that remains published.
- **Four Validation Layers:** XML format validation uses pysdmx 1.18.0 with validation enabled; SDMx-JSON is tested against the official 2.0.0 JSON Schema. The reviewed BIS LBS 1.0 snapshot supplies required component names, order and codelists. The reference profile also checks sender-country consistency, duplicates, supported periods and full-snapshot actions. Arithmetic checks are a separate fourth layer. Full dataflow/provision-agreement content constraints and formal standards certification are not implemented.
- **Three-Place Measures:** This platform's explicit convention is decimal half-up rounding once at intake to `DECIMAL(38,3)`, with three decimal places retained through calculation, storage and export. JSON uses decimal-aware serialization; API measure text avoids browser-number conversion. This is a reference profile, not a universal SDMx requirement. Original arrivals remain the evidence of pre-normalized input; zero and masked/missing are distinct.
- **Revision Without Regression:** A quarantined re-filing is written as a non-current audit record. The previously published version stays `IS_CURRENT` and continues to feed the public view, so a failed revision can never withdraw data that was already correct.

Standards-Native Dissemination

Data arrives as SDMx and leaves as SDMx. The governed result of a query is serialised back into the same dataflow it was reported against — `BIS:WS_LBS_D_PUB(1.0)` — rather than into a portal-specific export shape that a receiving system would have to learn.

The samples in demo/sdmx are historical administrator exports for `2026-Q1`, not current schema-conformance evidence. Current tests generate fresh messages with an explicit synthetic profile: `DECIMALS=3`, `UNIT_MEASURE=USD`, `UNIT_MULT=6`, `COLLECTION=E`, and frequency-derived `TIME_FORMAT`. `L_DENOM` classifies currency denomination; it does not change the observation's reporting unit. No currency conversion is performed.

These saved samples contain the published slice, not the 44-row quarantine-inclusive audit view. The JSON sample was regenerated locally from the same observations after a serializer correction; it is not an untouched browser capture. They demonstrate cross-format data equivalence, not independent verification of every persona or formal standards certification.

```
<Series FREQ="Q" L_MEASURE="S" L_POSITION="C" L_INSTR="A" L_DENOM="USD"
        L_CURR_TYPE="D" L_PARENT_CTY="5J" L_REP_BANK_TYPE="A" L_REP_CTY="US"
        L_CP_SECTOR="A" L_CP_COUNTRY="5J">
  <Obs TIME_PERIOD="2026-Q1" OBS_VALUE="400.000" OBS_STATUS="A" OBS_CONF="N" />
</Series>
```

Every dimension of the eleven-part key is written out, so the reporting jurisdiction the row filter keyed on (`L_REP_CTY="US"`, segment 9) is visible to the receiving system rather than implied. The SDMx-CSV rows carry the same structural identity on every line, which is what makes the file self-describing rather than order-dependent:

```
STRUCTURE,STRUCTURE_ID,ACTION,FREQ,...,TIME_PERIOD,OBS_VALUE,OBS_STATUS,OBS_CONF
dataflow,BIS:WS_LBS_D_PUB(1.0),I,Q,...,2026-Q1,400.000,A,N
```

Entitlement is enforced before serialization. The expected published-slice outcomes are listed below; the saved admin artifacts alone do not prove the other two personas. Downloaded files do not themselves enforce continuing access restrictions.

Persona	Observations exported	Restricted values
Public proxy	13	none present — confidential rows never enter the result
Researcher	13	none present — the 9 coordinate-masked rows stay in the portal as discovery metadata
Administrator	22	9 present, because `OBS_CONF = 'N'` is readable at this tier

The distinction between an absent observation and a zero one is not cosmetic. Writing `0` for a redacted value would convert a confidentiality control into a false data point that reconciles incorrectly downstream. Coordinate-masked rows never reach a download, the serializer refuses a masked key outright, and a masked measure is omitted rather than written as zero; `tests/test_sdmx_validation_rules.py` asserts both.

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6. Operational Playbook & Scale Strategy

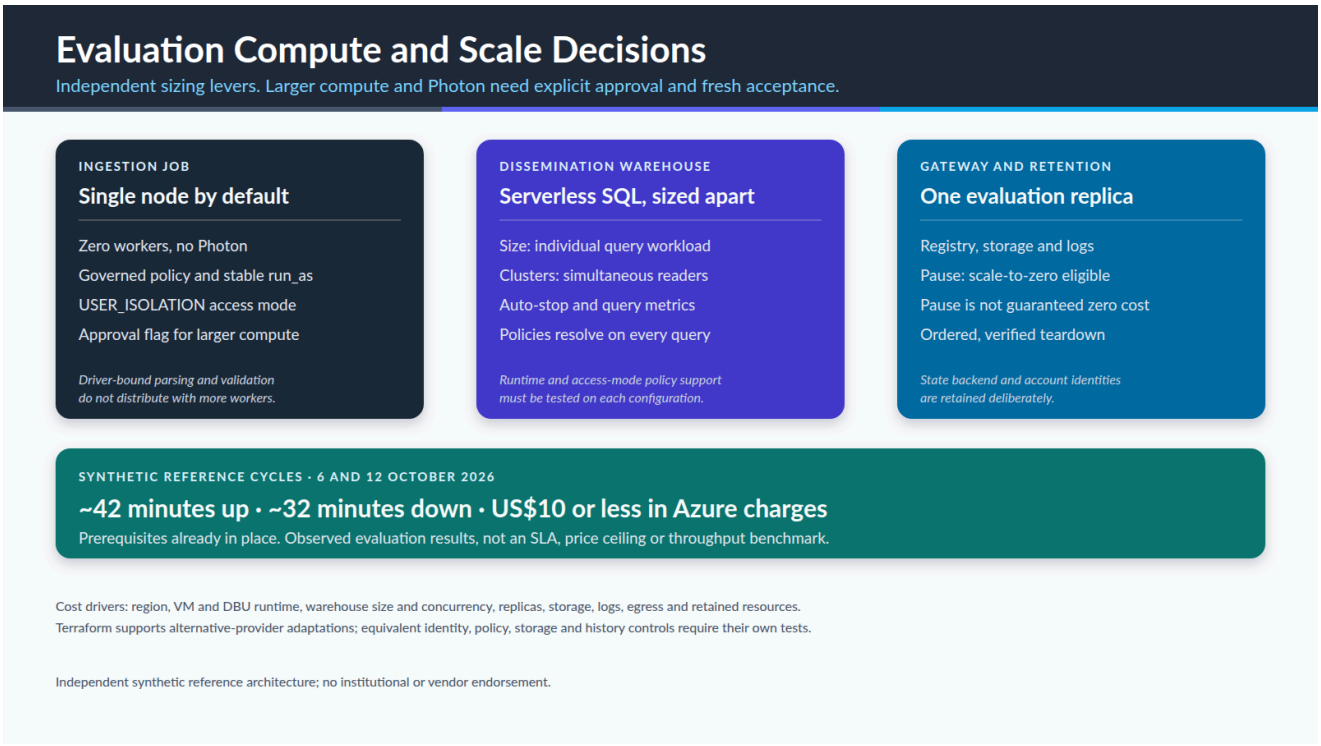


Figure 10 — Scalability & Compute Strategy: ingestion and dissemination compute can be sized independently while preserving the configured policy model.

The Minimal Viable Synthetic Dataset (MVSD) Protocol

An institution does not need to share internal records to start development:

1. The institution extracts structural metadata from its DSD or schema catalog.
2. The mock generator (`src/generate_sovereign_submissions.py`) produces an authentic synthetic fixture exercising all security branches: multiple jurisdictions, free-to-publish and confidential observations, and a revision cycle whose figures break named checks in the published BIS workbook. The corrupted submissions use only real, permitted codelist values — the failures are genuine arithmetic inconsistencies a validator detects, not malformed records a parser would reject, because a fixture that fails at parse time never reaches the controls it is meant to test.
3. External contractors build and validate all SQL, PySpark, and Terraform logic against the MVSD using local test harnesses.

Reproducible Deployment and Teardown

After the one-time remote-state backend and local configuration are prepared, `sh/sovereignshield_up.ps1` executes the validated sequence from Terraform foundation through Databricks account wiring, Asset Bundle deployment, ingestion, grants, both portal hosts, and readiness checks. Stages can be bounded or resumed after a cloud timeout. `sh/sovereignshield_down.ps1` deletes tables and policy functions before the schemas that contain them, destroys bundle and Azure resources through their owning paths, removes orphaned diagnostics, and verifies both empty Terraform state and an empty workload resource group. The backend and account-level identity records are deliberately retained for reliable reconstruction.

The 6 October 2026 reference evaluation took approximately **42 minutes to bring up the project, with prerequisite setup already in place**, and the complete deploy, test and teardown cycle incurred **US$10 or less in Azure charges**. The latest teardown (12 October 2026) took approximately **32 minutes**. This supports a low-cost, bounded synthetic evaluation; it is not a guaranteed ceiling for other regions, subscriptions, durations or workloads. Retained backend resources remain separately governed. The evaluation evidence defines measurement provenance and accounting limits.

The reason each Azure and Databricks object exists, who creates it, and whether it is reused or deleted is documented in the Systems Architect Resource Provenance Guide. The command sequence and recovery controls are in the One-Command Operations Runbook.

Cluster Sizing & Enterprise Scale

Single-node compute is an evaluation cost choice. Larger supported query engines must preserve the same entitlement contract, but runtime/access-mode support, performance and policy behavior require validation on each selected configuration.

- **Sandbox Evaluation:** The complete architecture runs on a single-node `USER_ISOLATION` cluster (`worker_count_max = 0`), keeping evaluation inexpensive for an organization or an external contractor assessing the controls.
- **Opt-In Compute:** Terraform now emits the job cluster specification, including its policy ID, and the bundle consumes it. Worker counts and Photon remain off by default and require explicit cost approval. Increasing workers does not distribute the pandas XML parsing and arithmetic-validation stages; end-to-end scale and efficiency require measurement and further engineering.
- **Dissemination:** Databricks SQL Serverless scales independently, by size (Medium through 2X-Large) for heavy individual scans and by `sql_warehouse_max_clusters` for concurrent readers. Concurrency and scan cost are separate levers, and a public dissemination tier usually needs the second one first.
- **Stress Test Verification:** The platform includes a scale harness (`src/generate_stress_test_data.py`) that produces a reproducible 100,000+ observation corpus spanning seven jurisdictions and all four reporting cadences, with configurable confidentiality and revision rates. `tests/test_scale_and_stress.py` asserts that entitlement enforcement stays vectorised and roughly linear in row count, and that SCD2 interval integrity holds at volume.

  > **Measurement note.** The published figures are the corpus shape and the linearity assertion, both reproducible offline. Latency under concurrent multi-user load on production-sized hardware has not been benchmarked here, and is not claimed.

Open Statistical Reconstruction Challenge

Masked values can be reconstructed from released totals, components, overlapping breakdowns or successive revisions. For example, the synthetic fixture permits a restricted component to be calculated as $1000-400-500=100$. Knowledge of the calculation method and the presence of a specific observation can make an otherwise hidden value identifiable. Researcher-visible keys, confidentiality flags and row counts therefore require information-disclosure review as well as value masking.

Community review is explicitly invited using synthetic examples: identify the released inputs, calculation, inferred value and affected persona. If observation existence makes reconstruction trivial, the Researcher role must be restricted or removed from a production design until an approved disclosure control is demonstrated. Disabling that role alone does not resolve reconstruction possible from public values. Secondary suppression, approved perturbation or a redesigned release product must be evaluated across dimensions, time and cumulative exports. The security challenge defines reporting and acceptance criteria.

The Secure Data Enclave (Conceptual Future State)

This section describes an architectural concept and policy mechanism. The repository contains no enclave infrastructure.

Two releases of the same cells, one suppressed for the public and one less suppressed for researchers, can be differenced to recover what either withholds. Sovereign Shield therefore applies **unified public suppression**: every product released without a value entitlement, to the public or to researchers, contains the same releasable cells. Submitters receive restricted cells only for their own filings, and administrators only under their oversight entitlement. The Discovery Gateway adds only the existence of restricted series, never their values or exact coordinates.

A researcher who needs restricted cells, for example to estimate a regression on confidential macro positions, applies to the originating authority for a **Secure Data Enclave**: a locked-down Unity Catalog workspace in which approved code runs against restricted data but raw cells cannot be exported. The enclave follows the widely used Five Safes model.

Safe	Enclave mechanism, conceptual
Projects	A named, time-bounded project approved by the submitting authority, with a stated statistical purpose
People	Membership of a project enclave group under a signed agreement; expiry removes it
Settings	A dedicated workspace bound to the restricted catalog, private networking with no outbound internet, result download and file export disabled, all activity in audit system tables
Data	A project-scoped dynamic view or attribute-based policy exposing only approved series and periods
Outputs	Coefficients and aggregates leave only through a reviewed output volume after disclosure checks; raw extracts are refused

Technical settings reduce exfiltration; they cannot prevent manual transcription, which agreements and output checking govern. Databricks Clean Rooms are an alternative where several authorities must analyse data jointly without sharing it. Each mechanism needs platform-specific design, testing and the data authority's approval before use.

Production Acceptance Boundaries

The deployment uses synthetic data and is not a production accreditation. Logical country segregation in a shared Canadian workspace is not physical country residency. RLS/DDM enforces entitlements, not complete statistical disclosure control: published totals in the fixture can reconstruct a masked component. Secondary suppression or another approved disclosure method is required before treating that risk as controlled. The gateway, privileged operators, exported files, and source archives remain trust boundaries.

Institutional threat modelling, privacy review, retention, independent rule semantics, penetration testing, disaster recovery, private networking, legacy policy migration, and production cost/throughput measurements remain required. The release evidence and migration guide records the reproducible methods, limitations, and promotion gates.

Conclusion

SovereignShield separates public dissemination from restricted statistical access through explicit information contracts and platform controls. The gateway selects identity and lifecycle scope, Unity Catalog enforces entitlements, SDMx validation determines acceptance, and submission-aware history preserves the current accepted publication. The Analyst View reconciles expected filings with the receiver's actual state. The Discovery Gateway lets researchers find restricted series without values or coordinates, unified public suppression keeps every researcher download identical to the public product, and the Secure Data Enclave describes how approved analysis of restricted cells could proceed. Synthetic-first delivery and client-owned runtime identities support implementation and handover, while disclosure control and institutional production approval remain explicit acceptance gates.

Appendix A: Policy Examples

These are logical templates from the policy source, not commands to replace a bound function manually. The executor gives functions content-addressed names; existing incompatible tables cause refusal before policy changes.

```
CREATE OR REPLACE FUNCTION fn_ddm_obs_conf_mask(
  obs_val DECIMAL(38,3), obs_conf STRING, l_rep_cty STRING
)
RETURNS DECIMAL(38,3)
RETURN CASE
  WHEN is_account_group_member('sg-sovereignshield-admin') THEN obs_val
  WHEN is_account_group_member('sg-sovereignshield-submitter-ca')
    AND coalesce(l_rep_cty = 'CA', FALSE) THEN obs_val
  WHEN is_account_group_member('sg-sovereignshield-submitter-us')
    AND coalesce(l_rep_cty = 'US', FALSE) THEN obs_val
  WHEN upper(trim(coalesce(obs_conf, ''))) = 'F' THEN obs_val
  ELSE NULL
END;

CREATE OR REPLACE FUNCTION fn_ddm_series_key_mask(
  time_series_code STRING, obs_conf STRING, l_rep_cty STRING
)
RETURNS STRING
RETURN CASE
  -- the same four reveal branches as fn_ddm_obs_conf_mask
  WHEN coalesce(size(split(time_series_code, '\\.')) = 11, FALSE)
    THEN concat(substring_index(time_series_code, '.', 9), '.xx.xx')
  ELSE NULL
END;
```

```
CREATE OR REPLACE FUNCTION fn_rls_multi_persona_lock(
  l_rep_cty STRING, batch_status STRING, obs_conf STRING
)
RETURNS BOOLEAN
RETURN
  is_account_group_member('sg-sovereignshield-admin')
  OR (
    is_account_group_member('sg-sovereignshield-researchers')
    AND upper(coalesce(batch_status, '')) = 'PUBLISHED'
  )
  OR (
    (is_account_group_member('sg-sovereignshield-public')
      OR is_account_group_member('sg-sovereignshield-submitter-ca')
      OR is_account_group_member('sg-sovereignshield-submitter-us'))
    AND upper(coalesce(batch_status, '')) = 'PUBLISHED'
    AND upper(trim(coalesce(obs_conf, ''))) = 'F'
  )
  OR (is_account_group_member('sg-sovereignshield-submitter-ca')
    AND coalesce(l_rep_cty = 'CA', FALSE))
  OR (is_account_group_member('sg-sovereignshield-submitter-us')
    AND coalesce(l_rep_cty = 'US', FALSE));
```

Appendix B: Context and References

The bounded contribution is an inspectable integration of synthetic-first specialist delivery, company-controlled runtime identity, table-level entitlement policy, coordinate-masked discovery, arrival-aware revision handling and separate dissemination/audit contracts. It does not claim global novelty, replace SDMx registries, or implement a complete statistical disclosure-control system.

The design reflects practical experience with group-based row-level security for regulatory returns on a commercial cloud platform, the modernization of an international banking statistics pipeline to SDMx 3.0 output, and the SDMx community's open tooling. It responds to three recurring needs: masking that no developer can bypass, governed partnerships between data holders and researchers, and analysts' need to know which of several submissions an international organization holds.

Existing approaches include SDMx Reference Infrastructure, Fusion Metadata Registry, and pysdmx. A registry maintains structural and provisioning metadata; the reference platform consumes a pinned subset and adds an Azure/Delta execution and access-control example. An institution may retain its current platform, extend existing SDMx tooling, or pilot this Azure integration after comparing operating burden, interoperability and residency requirements.

1. SDMx naming evolution, 29 June 2026. Display prose uses SDMx; protocol identifiers and published schema names retain their defined spelling.
2. pysdmx structural validation guidance. Format parsing, DSD requirements, codelists and provisioning constraints are distinct checks.
3. BIS LBS 1.0 DSD and referenced structures. Snapshot source and digest are recorded in the repository.
4. Unity Catalog dynamic views. Membership functions are caller-aware; underlying-object privileges are a separate concept.
5. Row filters and masks. Runtime and access-mode limitations must be checked for the deployed engine.
6. Terraform sensitive data. Sensitive marking does not remove values from state or plan files.
7. BIS terms of use. Public availability is not a blanket redistribution licence; third-party artifacts are excluded from this project's Apache grant.
8. Attribute-based access control in Unity Catalog. Tag-driven policies are the platform's recommended way to apply the same filters and masks across many tables.

9. ONS Secure Research Service. A national statistics office applying the Five Safes model to secure research access.